Product Requirements Document (PRD)

Codebase Cartograph — *Map your codebase. Understand everything.*

> **Status:** Draft v1.0
> **Date:** March 22, 2026
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Table of Contents

1. Executive Summary
2. Problem Statement
3. Vision & Goals
4. Target Users
5. System Architecture
6. Core Engine (CLI)
7. Web UI
8. IDE Extensions
9. Coding Agent Integration
10. Data Model
11. API Contracts
12. Auto-Update System
13. Configuration
14. Cost Analysis
15. Build Phases & Roadmap
16. Tech Stack
17. Non-Functional Requirements

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1. Executive Summary

Codebase Cartograph is an **open-source, AI-powered codebase intelligence platform** that transforms any codebase into a navigable, queryable knowledge system. It combines static analysis, LLM-powered documentation generation, knowledge graph construction, and a conversational interface to let both humans and AI agents deeply understand codebases.

Core value proposition: Instead of reading thousands of files to understand a codebase, users interact with a structured documentation pyramid, an interactive knowledge graph, and a chat interface that routes questions to the right level of detail automatically.

Key differentiators: - **Hierarchical documentation pyramid** — file → directory → module → architecture summaries - **Knowledge graph** — not just docs, but a queryable map of how everything connects - **"Chat with your codebase"** — RAG-powered conversational interface with visual context - **Coding agent integration** — hand off tasks to AI agents with perfect context - **IDE-native** — VS Code, Cursor, and PyCharm extensions - **Fully configurable** — any LLM provider, any model, any codebase - **Self-contained** — runs on a laptop or deploys for a team

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2. Problem Statement

The Challenge

Modern codebases are large, complex, and poorly documented. The Salt AI `ai-studio` repository alone contains **977 Python files across 158K lines of code** spanning microservices, SDKs, node systems, and infrastructure tooling.

Current Pain Points

Pain Point	Impact
Onboarding takes weeks	New developers spend days reading code before contributing
LLMs lack context	AI coding assistants choke on large repos — too many files for context windows
Documentation is stale	Manual docs drift from reality within days of being written
	How modules connect lives in senior devs' heads, not in docs

Pain Point	Impact
Architecture is tribal knowledge	
Code review lacks context	Reviewers don't know the full impact of changes
No "find similar"	Developers reinvent patterns that already exist in the codebase

The Cost

- **Developer time**: ~30% of engineering time spent understanding existing code (industry average)
- **AI inefficiency**: Coding agents produce wrong/inconsistent code without proper context
- **Knowledge loss**: When senior developers leave, architectural understanding leaves with them

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3. Vision & Goals

Vision

A world where every codebase is immediately understandable — where a new developer, an AI agent, or a curious PM can ask any question about any codebase and get an accurate, contextual answer in seconds.

Goals

Goal	Metric
Reduce onboarding time	New dev productive in days, not weeks
Improve AI coding accuracy	Agents produce codebase-consistent code on first attempt
Keep docs perpetually fresh	Auto-update within minutes of code changes
Make architecture visible	Interactive graph accessible to all stakeholders
Enable self-service understanding	Any question answered without bothering a senior dev

Non-Goals (v1)

- Not a code editor or IDE replacement
- Not a CI/CD tool
- Not a code review system (though it supports reviews)
- Not language-specific (Python first, extensible to others)

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4. Target Users

Primary Users

Software Engineers — Understand unfamiliar code, navigate dependencies, find patterns, get context for changes.

AI Coding Agents — Consume the documentation pyramid and knowledge graph as context for generating accurate, consistent code.

Engineering Managers / Tech Leads — Understand architecture, assess impact of changes, identify tech debt and complexity hotspots.

Secondary Users

Product Managers — Understand what systems do at a high level without reading code.

QA Engineers — Understand test coverage, identify untested paths, trace data flow.

New Hires — Self-service onboarding with guided exploration of the codebase.

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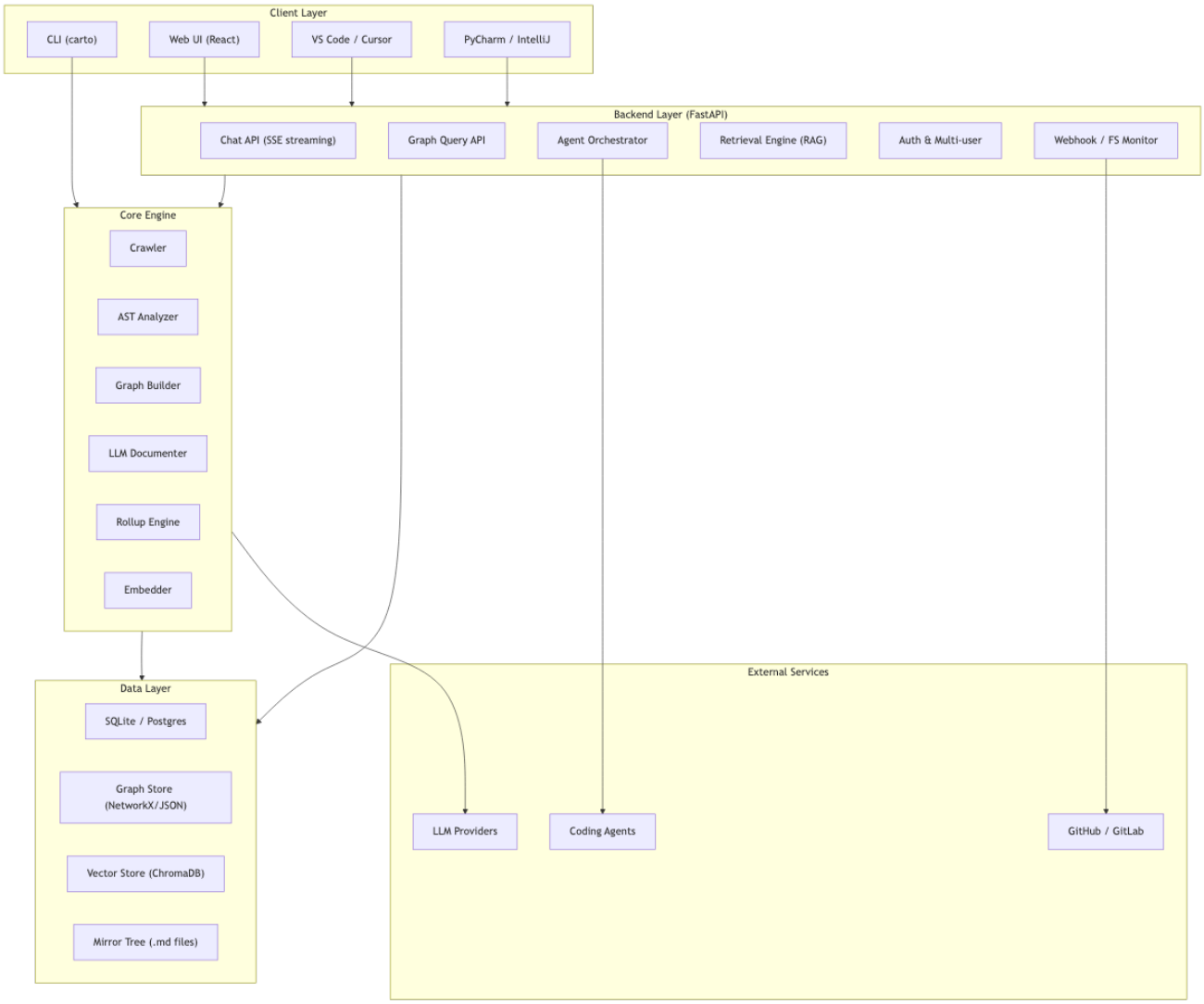
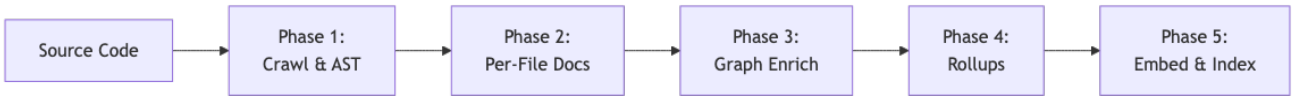
5. System Architecture

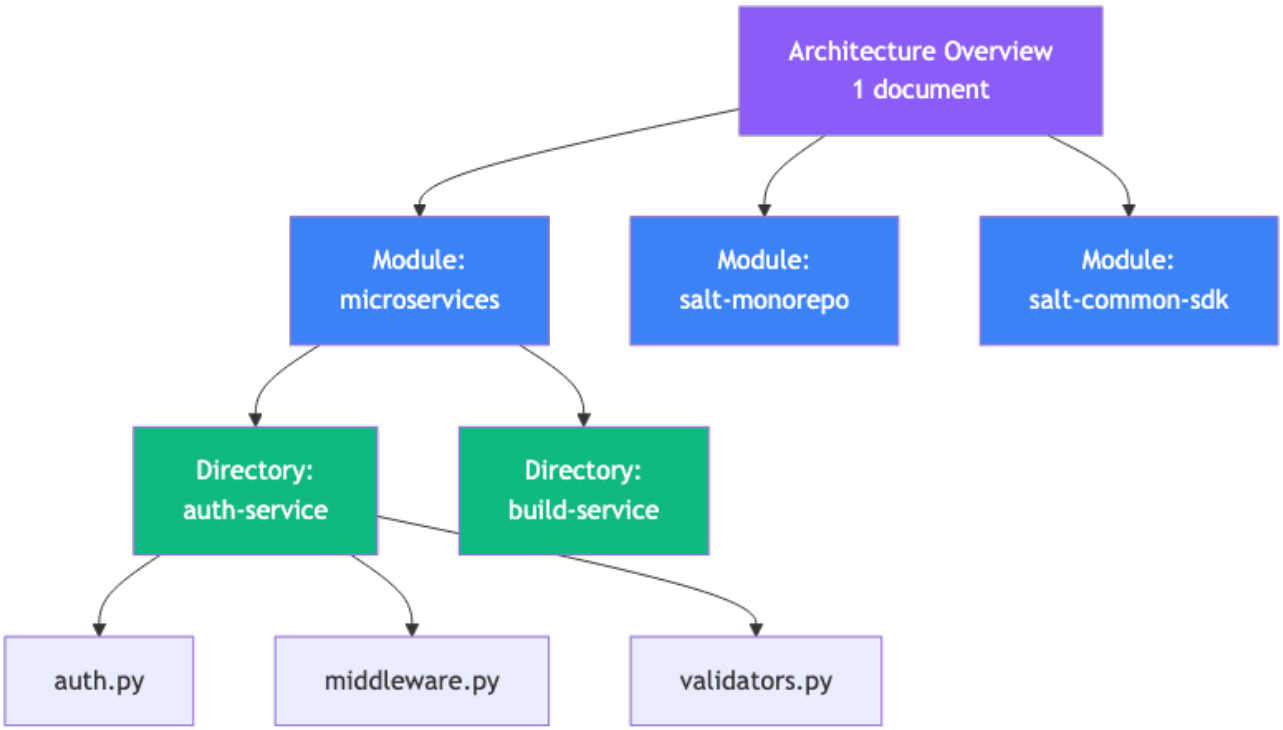
High-Level Architecture



Processing Pipeline



Documentation Pyramid



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6. Core Engine (CLI)

Commands

Command	Description	LLM Required
`carto analyze <path>`	AST parse + build knowledge graph	❌ Free
`carto document <path>`	Generate per-file .md documentation	✅
`carto enrich <path>`	Annotate graph edges with semantic meaning	✅
`carto rollup <path>`	Generate hierarchical summaries	✅
`carto run <path>`	All phases in sequence	✅

Command	Description	LLM Required
`carto chat <path>`	Interactive chat REPL	✅
`carto context <path> <query>`	Export relevant context for external use	✅
`carto stats <path>`	Show codebase stats + cost estimates	❌ Free
`carto serve`	Launch web UI + API server	❌

Phase 1: Static Analysis (AST)

Input: Source code files
Output: Knowledge graph (JSON) + analysis cache

Extracts: - **Imports** — internal vs external, aliased imports, star imports - **Functions** — signatures, args with type hints, return types, decorators, docstrings, complexity score - **Classes** — bases, methods, class variables, decorators, metaclasses - **Call graph** — which functions call which other functions (cross-file) - **Module variables** — constants, configuration, type aliases - **Dependency edges** — typed relationships between all entities

Knowledge graph edge types:

Edge Type	Meaning	Example
`imports`	Module imports another	`auth.py → jwt_utils.py`
`calls`	Function calls another function	`validate_user() → check_token()`
`inherits`	Class extends another	`AdminUser → BaseUser`
`composes`	Class holds reference to another	`UserService has DBConnection`
`decorates`	Decorator applied to function/class	`@require_auth → upload_handler`
`uses`	Variable/type reference	`handler uses UserSchema`

Phase 2: Per-File Documentation

Input: Source file + AST analysis + local graph context
Output: Structured markdown document

Document structure:

```
# auth.py

## Purpose
One-paragraph description of what this file does and why it exists.

## Dependencies
- **Internal:** jwt_utils.py (token validation), user_model.py (data types)
- **External:** fastapi, pydantic, jose

## Classes

### AuthMiddleware
Base: `BaseHTTPMiddleware`
Purpose: Intercepts requests to validate JWT tokens.
Methods:
- `dispatch(request, call_next)` — Main middleware handler
- `_extract_token(request)` — Pulls JWT from Authorization header

## Functions

### validate_token(token: str) → TokenPayload
Validates and decodes a JWT token. Raises `AuthError` if expired or invalid.
Called by: `AuthMiddleware.dispatch`, `websocket_auth`
Calls: `jose.jwt.decode`, `TokenPayload.model_validate`

### require_auth(scopes: list[str]) → Callable
Decorator factory that creates auth dependency with required scopes.
Used by: 14 route handlers across 6 files

## Data Flow
Request → AuthMiddleware.dispatch → _extract_token → validate_token → route handler

## Side Effects
- Reads JWT_SECRET from environment variables
- Logs authentication failures to structured logger
- Raises HTTP 401/403 on auth failure
```

Phase 3: Graph Enrichment

Takes the raw AST-extracted graph and adds semantic meaning via LLM: - **Why** does A call B (not just that it does) - **Data flow** annotations on edges (what data passes between) - **Pattern identification** (this is a factory, this is a middleware chain, etc.)

Phase 4: Hierarchical Rollup

Bottom-up summarization: 1. **Leaf directories** — summarize all file docs in a directory 2. **Parent directories** — summarize child directory summaries 3. **Module level** — summarize top-level project directories 4. **Architecture level** — one document summarizing the entire codebase

Each level captures not just *what's in* the children, but *how they relate to each other*.

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7. Web UI

Layout

```
┌──────────────────────────────────────────────────────────────────────┐
│  🧭 [Project ▼] [Scope: module/auth ▼]            [⚙️] [👤]    │
├──────────────────────────────┬──────────────────────────────────────┤
│                              │  [Source] [Graph] [Docs]             │
│    💬 CHAT PANEL             │  [Dependencies] [Examples]           │
│    (60% width, resizable)    │                                      │
│                              │    CONTEXT PANEL                     │
│    Streaming conversation    │    (40% width, tabbed)               │
│    with clickable refs       │                                      │
│    and action buttons        │    Updates dynamically               │
│                              │    as chat responds                  │
│    ┌────────────────────┐    │                                      │
│    │ [📌 pinned files]   │    │                                      │
│    │ [/trace] [@file] [+]│    │                                      │
│    │ [Message input...  ]│    │  ┌──────────────────────────┐        │
│    └────────────────────┘    │  │ Token: 34K/128K │ 💰$0.03  │        │
├──────────────────────────────┴──┴──────────────────────────────────┤
│  📊 977 files indexed │ 4.2K graph nodes │ Last updated: 2m ago │
└──────────────────────────────────────────────────────────────────────┘
```

Chat Panel Features

Feature	Description
Streaming responses	Token-by-token SSE rendering
Clickable references	`auth.py:42` links navigate the context panel

Feature	Description
Action buttons	🔍 Trace, 🔧 Fix This, 📋 Export, 🔄 Regenerate
Slash commands	`/trace fn`, `/impact file`, `/similar pattern`, `/scope path`
File autocomplete	`@auth.py` — autocomplete file/function names
Pin context	`+auth.py` keeps files in persistent context
Depth control	`/depth architecture | module | file | source`
Conversation branches	Fork at any message for "what if" exploration
Suggested follow-ups	3-4 contextual next questions after each answer
Conversation history	Saved per-project, searchable

Context Panel Tabs

Tab 1: Source View

- **Monaco editor** (read-only) — VS Code's renderer
- Multi-file tabs within the panel
- **Highlighted regions** that chat is referencing (yellow glow animation)
- Gutter annotations: complexity, dependency count, test coverage
- **"Open in IDE"** button → launches VS Code/Cursor/PyCharm at exact file:line
- Diff mode toggle for git changes

Tab 2: Knowledge Graph (Interactive)

- **Cytoscape.js** force-directed graph visualization
- **Semantic zoom levels:**
- Zoomed out → module clusters (colored bubbles)
- Mid zoom → file nodes within modules
- Zoomed in → class/function nodes within files
- **Edge type visualization:**

Edge Type	Visual Style
Imports	Dashed gray line

Edge Type	Visual Style
Function calls	Solid blue arrow
Inheritance	Thick green line
Composition	Dotted orange line
Decorates	Purple line

- **Live highlighting**: nodes pulse/glow as chat discusses them
- **Click interactions**: click node → summary tooltip + source navigation
- **Filters**: toggle edge types, filter by module, search nodes
- **Layout modes**: force-directed, hierarchical, circular, tree
- **Export**: SVG/PNG snapshot of current view

Tab 3: Documentation Pyramid

- Collapsible tree view (file explorer style)
- Level icons: 🏗️ architecture, 📦 module, 📁 directory, 📄 file
- Click any node → renders markdown in-place
- Staleness indicator: ⚠️ when source changed since generation
- Search within docs
- Team annotations with author + timestamp

Tab 4: Dependencies

- Two-column: **Upstream** | **Downstream**
- Visual DAG (pipeline diagram)
- External deps link to PyPI/docs
- **Impact radius**: "change this → N files affected"
- Expandable: click a dep → see which functions/classes are used

Tab 5: Examples & Patterns

- Similar code across the codebase
- Test examples for the current topic
- Usage examples (how other code calls this)
- Pattern classification: "Repository Pattern", "Factory", "Middleware Chain"

Interactive Features

Feature	Description
Trace	Click a function → animated call trace through the graph
Impact Analysis	"What if I changed this?" → highlight cascade in graph
Find Similar	Right-click code → find other places using the same pattern
What-If Mode	"What if I deleted this class?" → show everything that breaks
Comparison	Side-by-side module comparison (graph + docs)
Explain Like I'm...	Toggle depth: junior / senior / PM / AI agent
Annotations	Team notes on any doc/node that persist and accumulate
Pin Context	Pin files/functions to always be in chat context
Export Conversation	Chat + context panel → shareable document
Git-Aware	"What changed in last 5 commits affecting this module?"

---

8. IDE Extensions

Architecture

IDE extensions are **thin clients** that call the Cartograph API. No engine logic in the extension — just UI.

```
graph LR
    subgraph IDE["IDE (VS Code / Cursor / PyCharm)"]
        Sidebar["Sidebar: Mini Graph"]
        Hover["Hover: Doc Summary"]
        RightClick["Context Menu Actions"]
        ChatPanel["Chat Panel"]
        Commands["Command Palette"]
    end

    subgraph Server["Cartograph Backend"]
```

```
        API["FastAPI Server"]
    end

    Sidebar -->|"GET /graph/subgraph"| API
    Hover -->|"GET /files/:id/doc"| API
    RightClick -->|"POST /context"| API
    ChatPanel -->|"POST /conversations/:id/messages (SSE)"| API
    Commands -->|"Various endpoints"| API
```

VS Code / Cursor Extension

Feature	Description
Sidebar panel	Mini knowledge graph for current file
Hover enrichment	Hover function → Cartograph doc summary + callers/callees
"Explain this"	Right-click → opens chat focused on selection
"Impact of change"	Right-click function → see affected files
"Find similar"	Right-click pattern → find matches across codebase
Inline annotations	Subtle decorations: dependency count, complexity
Chat panel	Full chat interface embedded in editor
Command palette	`Cartograph: Trace`, `Cartograph: Dependencies`, `Cartograph: Explain`
Status bar	Connection status, last index time, file doc staleness

PyCharm / IntelliJ Plugin

Same features, adapted to JetBrains platform: - Tool window instead of sidebar - Intention actions instead of right-click menu - Built in Kotlin (required by JetBrains API) - Same backend API

Priority: VS Code/Cursor first (same codebase), PyCharm second.

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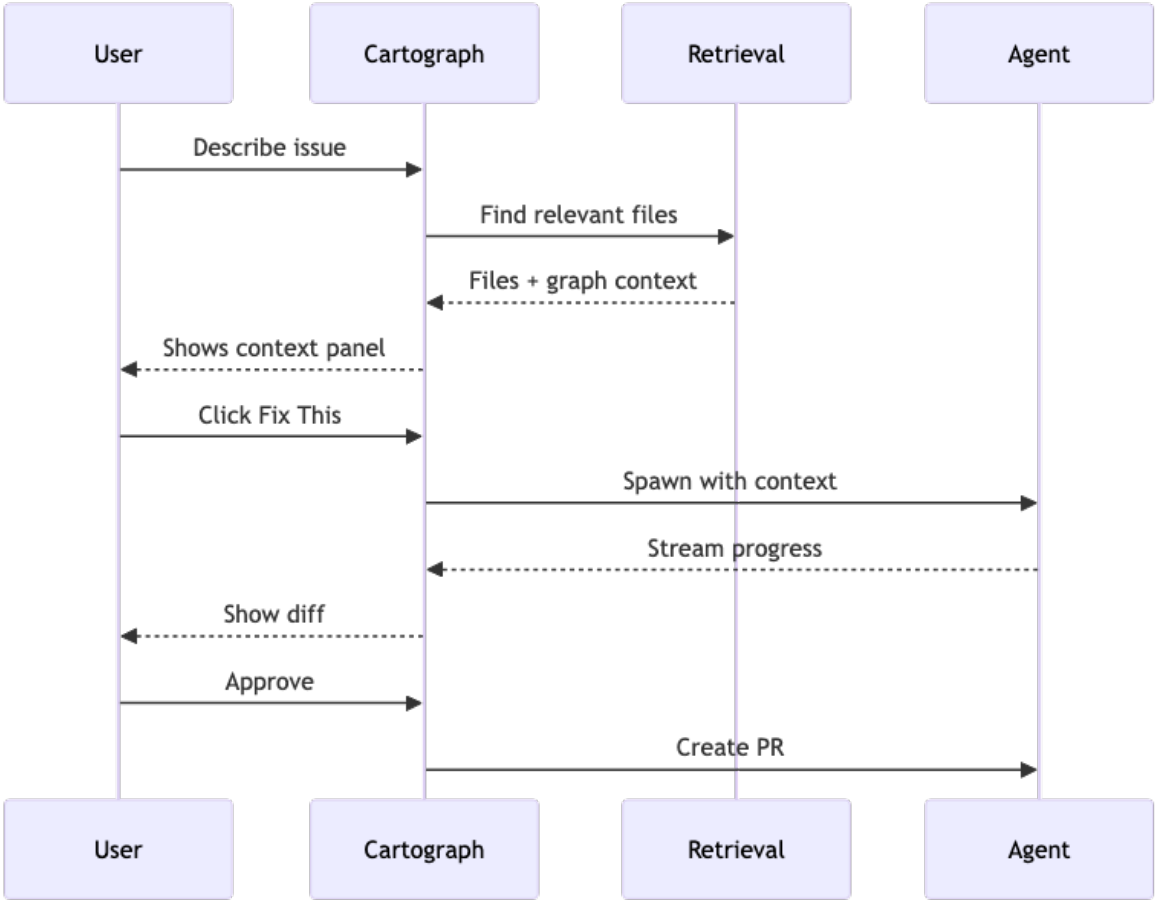
9. Coding Agent Integration

The Killer Feature

Cartograph doesn't just document code — it becomes an **AI coding accelerator** by providing perfect context to coding agents.

"Fix This" Flow



"Implement Feature" Flow

```
sequenceDiagram
    participant U as User
    participant C as Cartograph
    participant G as Knowledge Graph
    participant A as Coding Agent
```

```
    U->>C: "Add rate limiting to all API endpoints"
    C->>G: Query: find all API endpoints
    G-->>C: 23 endpoints across 8 files
    C-->>U: Shows endpoint list, user selects/deselects
    U->>C: Clicks "Implement"
    C->>A: Spawn with endpoint list + decorator patterns + middleware arch
    A-->>C: Implements across all files
    C-->>U: Shows multi-file diff
```

"Review PR" Flow

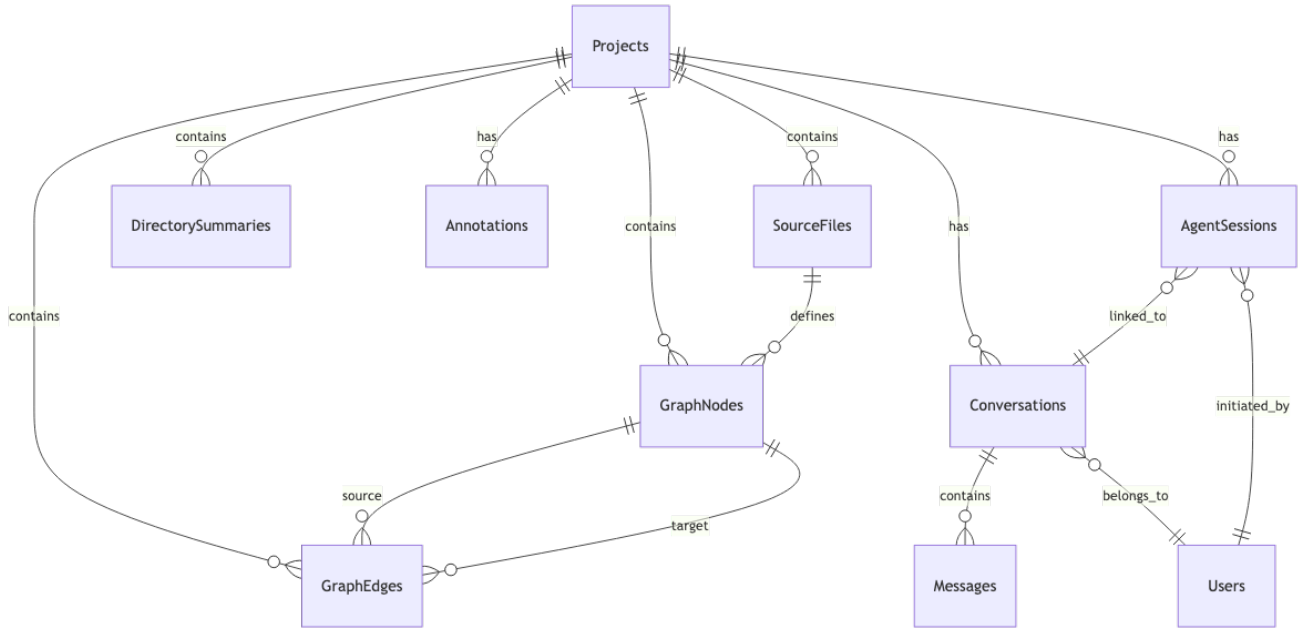
```
sequenceDiagram
    participant GH as GitHub
    participant W as Webhook
    participant C as Cartograph
    participant G as Knowledge Graph

    GH->>W: Push event (new PR)
    W->>C: Trigger PR analysis
    C->>G: Map changed functions to graph
    G-->>C: Downstream impacts, missing tests, stale docs
    C-->>GH: Post review summary to PR
    C-->>U: Available in chat: "Explain this PR"
```

Agent Configuration

Setting	Options
Agent type	Codex, Claude Code, custom command
Model per agent	Configurable in settings UI
Working directory	Auto-creates feature branch
Branch strategy	`fix/<task>`, `feature/<task>`, custom pattern
Auto-PR	Automatic or manual review
Sandboxing	Full sandbox, workspace-only, or unrestricted

10. Data Model

Entity Relationship Diagram



11. API Contracts

Project Management

```
GET    /api/projects                          → List all projects
POST   /api/projects                          → Create project
GET    /api/projects/:id                      → Project details + stats
PATCH  /api/projects/:id                      → Update config
DELETE /api/projects/:id                      → Remove project
POST   /api/projects/:id/index                → Full re-index
POST   /api/projects/:id/index/incremental    → Incremental update
```

Graph & Source

```
GET    /api/projects/:id/graph                → Full graph (JSON)
GET    /api/projects/:id/graph/subgraph       → Local subgraph
```

```
          ?node=X&depth=2
GET    /api/projects/:id/graph/nodes              → Search nodes
          ?type=function&search=auth
GET    /api/projects/:id/graph/node/:nodeId       → Node details + edges
GET    /api/projects/:id/graph/trace/:nodeId      → Full call trace
GET    /api/projects/:id/graph/impact/:nodeId     → Impact analysis
GET    /api/projects/:id/graph/similar/:nodeId    → Similar patterns
GET    /api/projects/:id/graph/mermaid?scope=path → Mermaid diagram

GET    /api/projects/:id/files                    → File listing
GET    /api/projects/:id/files/:fileId            → File details
GET    /api/projects/:id/files/:fileId/source     → Raw source code
GET    /api/projects/:id/files/:fileId/doc        → Generated documentation
```

Documentation Pyramid

```
GET    /api/projects/:id/docs/tree                → Doc tree structure
GET    /api/projects/:id/docs/:path               → Doc at path
GET    /api/projects/:id/docs/architecture        → Top-level doc
GET    /api/projects/:id/docs/search?q=auth       → Search docs
```

Chat (Streaming)

```
GET    /api/projects/:id/conversations            → List conversations
POST   /api/projects/:id/conversations            → New conversation
GET    /api/conversations/:convId/messages         → Message history
POST   /api/conversations/:convId/messages         → Send message
DELETE /api/conversations/:convId                  → Delete conversation
POST   /api/conversations/:convId/messages/:msgId/branch → Branch
```

Chat SSE Event Types:

```
{"type": "token", "content": "The"}
{"type": "reference", "file": "auth.py", "line": 42, "node_id": "..."}
{"type": "graph_highlight", "nodes": ["node1", "node2"], "edges": ["edge1"]}
{"type": "context_used", "docs": [...], "tokens": 6200}
{"type": "done", "cost": 0.003, "model": "gemini-2.5-pro"}
{"type": "suggestions", "questions": ["How does...", "What if..."]}
```

Context Export

```
POST   /api/projects/:id/context
  Request:  {query, depth, max_tokens}
  Response: {docs, graph, source_snippets, total_tokens}
```

Annotations

```
GET    /api/projects/:id/annotations?target=file:auth.py
POST   /api/projects/:id/annotations
DELETE /api/annotations/:id
```

Agent Integration

```
POST   /api/projects/:id/agents/run              → Spawn agent
GET    /api/agents/:sessionId                     → Status + logs
POST   /api/agents/:sessionId/approve             → Approve changes
DELETE /api/agents/:sessionId                      → Kill agent
```

Webhooks & Auto-Update

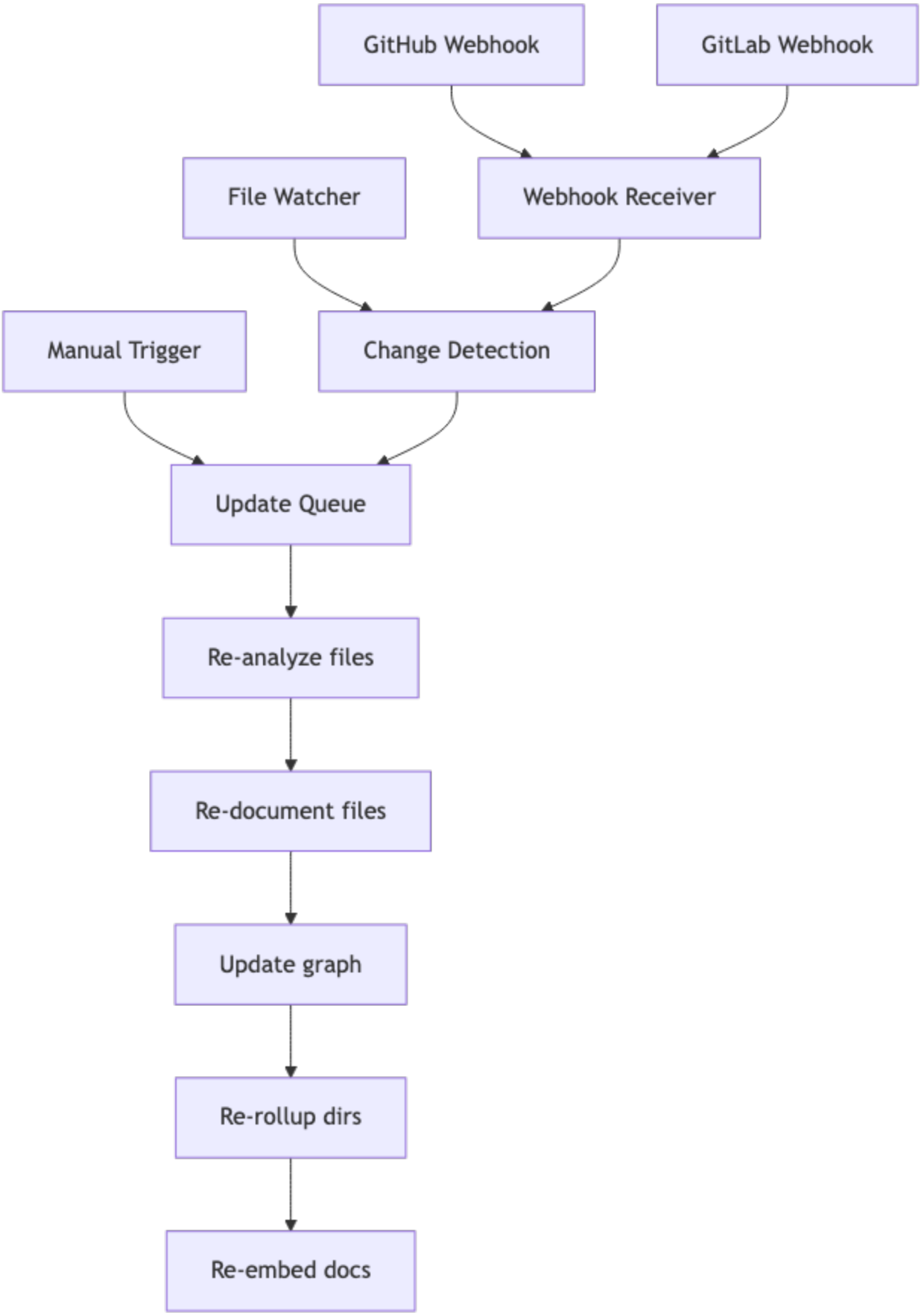
```
POST   /api/hooks/github                          → GitHub webhook
POST   /api/hooks/gitlab                          → GitLab webhook
GET    /api/projects/:id/watcher                  → Watcher status
PATCH  /api/projects/:id/watcher                  → Configure watcher
GET    /api/projects/:id/updates                  → Update log
```

Settings & Admin

```
GET    /api/settings                              → Current config
PATCH  /api/settings                              → Update config
GET    /api/settings/models                       → Available models
GET    /api/usage                                 → Usage/cost stats
GET    /api/users                                 → User management
```

12. Auto-Update System

Update Triggers


GitHub Webhook
GitLab Webhook
File Watcher
Webhook Receiver
Manual Trigger
Change Detection
Update Queue
Re-analyze files
Re-document files
Update graph
Re-rollup dirs
Re-embed docs

Configuration (UI Settings Panel)

Setting	Options	Default
Update method	Webhook / File watcher / Manual only	Manual
Webhook URL	Auto-generated, copyable	—
Watch paths	Include/exclude glob patterns	All source files
Debounce interval	Seconds to wait before processing	5s
Auto-rollup	Re-rollup affected directories on change	Yes
Notification	Alert in UI when update completes	Yes

---

13. Configuration

Configuration File (`config.yaml`)

```
# ============================================================
# Codebase Cartograph Configuration
# Copy to config.yaml and fill in your API keys
# ============================================================

# --- API Keys ---
api_keys:
  openai: ""          # sk-... (for embeddings, GPT models)
  anthropic: ""       # sk-ant-... (for Claude models)
  google: ""          # AIza... (for Gemini models)
  deepseek: ""        # (for DeepSeek models)

# --- Model Selection (per phase) ---
# Format: provider/model-name (via litellm)
models:
  file_documentation: "gemini/gemini-2.5-flash"
  graph_enrichment: "gemini/gemini-2.5-flash"
  rollup_summaries: "gemini/gemini-2.5-pro"
  chat: "anthropic/claude-sonnet-4-20250514"
  embeddings: "openai/text-embedding-3-small"

# --- Processing ---
```

```yaml
processing:
  concurrency: 10                 # Parallel LLM calls
  skip_tiny_files: true           # Skip files ≤ threshold
  tiny_file_threshold: 10         # Lines
  file_extensions: [".py"]        # Extensible: .ts, .js, .go, etc.
  exclude_patterns:
    - "**/test_*"
    - "**/.git/**"
    - "**/migrations/**"
    - "**/__pycache__/**"
    - "**/venv/**"
    - "**/.venv/**"
    - "**/node_modules/**"

# --- Output ---
output:
  output_dir: "./docs-mirror"
  graph_format: "json"            # json, sqlite, or both
  generate_mermaid: true
  include_source_links: true

# --- Chat ---
chat:
  retrieval_method: "keyword"     # "keyword" or "vector"
  top_k: 10
  include_source: true
  history_length: 10

# --- Server ---
server:
  host: "0.0.0.0"
  port: 8420
  mode: "solo"                    # "solo" or "team"
  auth_method: "none"             # "none", "api_key", "oauth"

# --- Auto-Update ---
updates:
  method: "manual"                # "manual", "webhook", "watcher"
  debounce_seconds: 5
  auto_rollup: true
```

Settings UI

All configuration is also editable through the web UI settings panel:

```
┌──────────────────────────────────────────┐
│  ⚙️ Settings                              │
│                                          │
│  📁 Projects                              │
│  ├ Add/remove codebases                  │
│  ├ Re-index / rebuild all                │
│  └ Per-project model overrides           │
│                                          │
│  🤖 Models                                │
│  ├ Model per phase (dropdown + test button) │
│  └ API keys (encrypted, masked display)  │
│                                          │
│  🔄 Auto-Update                           │
│  ├ Method: webhook / file watcher / manual  │
│  ├ Webhook URL (copy button + setup guide)  │
│  ├ Watch paths (include/exclude)         │
│  └ Update log (history of changes)       │
│                                          │
│  🛠️ Coding Agents                         │
│  ├ Available agents (auto-detected)      │
│  ├ Default agent per task type           │
│  └ Branch naming / PR settings           │
│                                          │
│  👥 Team (team mode only)                 │
│  ├ Users & roles                         │
│  ├ SSO / OAuth configuration             │
│  └ Shared annotation settings            │
│                                          │
│  💰 Usage Dashboard                       │
│  ├ Token usage by phase (chart)          │
│  ├ Cost over time (chart)                │
│  └ Budget alerts                         │
└──────────────────────────────────────────┘
```

---

14. Cost Analysis

Reference Codebase: Salt AI ai-studio

Metric	Value
Python files	977

Metric	Value
Lines of code	158,000
Total size	5.8 MB
Estimated tokens	~1.5M input
Directories	1,467
Max depth	15 levels
Top-level projects	11

File Size Distribution

Category	Count	Strategy
≤10 lines (tiny)	144	Skip (index only)
11-50 lines (small)	98	Cheap to process
51-200 lines (medium)	439	Bulk of the work
201-500 lines (large)	258	Substantial files
500+ lines (huge)	38	Largest: 2,379 lines

Cost by Model Strategy

Budget Strategy (~$5-7)

Phase	Model	Cost
Phase 1: AST Analysis	Local (free)	$0.00
Phase 2: Per-file docs	Gemini 2.5 Flash	$1.40
Phase 3: Graph enrichment	Gemini 2.5 Flash	$0.30
Phase 4: Rollups	Gemini 2.5 Pro	$3.00
Phase 5: Embeddings	text-embedding-3-small	$0.05

Phase	Model	Cost
Total		**~$4.75**

Premium Strategy (~$34)

Phase	Model	Cost
Phase 1: AST Analysis	Local (free)	$0.00
Phase 2: Per-file docs	Gemini 2.5 Pro	$26.00
Phase 3: Graph enrichment	Gemini 2.5 Pro	$5.00
Phase 4: Rollups	Gemini 2.5 Pro	$3.00
Phase 5: Embeddings	text-embedding-3-small	$0.05
Total		**~$34.05**

Maximum Quality (~$233)

Phase	Model	Cost
Phase 1: AST Analysis	Local (free)	$0.00
Phase 2: Per-file docs	Claude Opus 4	$173.00
Phase 3: Graph enrichment	Claude Opus 4	$35.00
Phase 4: Rollups	Claude Opus 4	$25.00
Phase 5: Embeddings	text-embedding-3-small	$0.05
Total		**~$233.05**

Incremental Update Costs

After initial indexing, updates only re-process changed files. Typical cost for a 10-file change: **$0.02-0.50** depending on model choice.

---

15. Build Phases & Roadmap

```
gantt
    title Codebase Cartograph — Build Roadmap
    dateFormat YYYY-MM-DD
    axisFormat %b %d

    section v0.1 Core Engine
    AST Analyzer & Graph          :done, v01a, 2026-03-22, 1d
    LLM Documenter               :done, v01b, 2026-03-22, 1d
    Rollup Engine                 :done, v01c, 2026-03-22, 1d
    CLI & Config                  :done, v01d, 2026-03-22, 1d

    section v0.2 Web UI + Chat
    FastAPI Backend               :v02a, 2026-03-24, 5d
    React Frontend (layout)       :v02b, 2026-03-26, 5d
    Chat streaming (SSE)          :v02c, 2026-03-28, 3d
    Source viewer (Monaco)        :v02d, 2026-03-29, 3d
    Graph visualization           :v02e, 2026-03-30, 5d
    Doc pyramid tree              :v02f, 2026-04-01, 2d

    section v0.3 Interactive
    Trace & Impact analysis       :v03a, 2026-04-05, 4d
    Find similar / What-if        :v03b, 2026-04-07, 3d
    Annotations system            :v03c, 2026-04-09, 3d
    Conversation branches         :v03d, 2026-04-10, 2d
    Pin & context controls        :v03e, 2026-04-11, 2d

    section v0.4 Auto-Update + Multi-User
    Webhook receivers             :v04a, 2026-04-14, 3d
    File watcher (watchdog)       :v04b, 2026-04-15, 2d
    Incremental pipeline          :v04c, 2026-04-16, 3d
    Auth layer & team mode        :v04d, 2026-04-18, 4d
    Settings UI                   :v04e, 2026-04-20, 3d

    section v0.5 IDE Extensions
    VS Code extension             :v05a, 2026-04-24, 7d
    Cursor support (fork test)    :v05b, 2026-04-29, 2d
    PyCharm plugin                :v05c, 2026-05-01, 7d

    section v0.6 Agent Integration
    Agent orchestrator            :v06a, 2026-05-08, 5d
    Fix This / Implement flows    :v06b, 2026-05-11, 4d
    PR integration                :v06c, 2026-05-14, 3d

    section v1.0 Release
```

```
    Docker deployment             :v10a, 2026-05-18, 3d
    Documentation & tutorials     :v10b, 2026-05-19, 4d
    PyPI + marketplace publish    :v10c, 2026-05-22, 2d
    Launch                        :milestone, v10d, 2026-05-24, 0d
```

Phase Details

Version	Target	Key Deliverables
v0.1 ✅	Mar 22	Core engine, CLI, AST analyzer, graph, LLM docs, rollups
v0.2	Apr 4	Web UI, chat interface, source viewer, graph viz
v0.3	Apr 13	Trace, impact, find similar, annotations, branches
v0.4	Apr 23	Webhooks, file watcher, auth, team mode, settings UI
v0.5	May 7	VS Code, Cursor, PyCharm extensions
v0.6	May 17	Agent orchestrator, Fix This, PR integration
v1.0	May 24	Docker, docs, PyPI, marketplace, launch

---

16. Tech Stack

Backend

Component	Technology	Rationale
Language	Python 3.11+	Same ecosystem as target codebases
Web framework	FastAPI	Async, fast, OpenAPI docs built-in
LLM client	litellm	Unified interface for all providers
Graph engine	NetworkX	Pure Python, batteries included
Database	SQLite (solo) / Postgres (team)	Zero-config vs production-ready
Vector store	ChromaDB (local) / Pinecone (cloud)	Embedded vs scalable

Component	Technology	Rationale
AST parsing	Python `ast` module	Standard library, deterministic
Task queue	asyncio + semaphore	No external deps for solo mode
File watching	watchdog	Cross-platform, mature
CLI	Click	Clean, composable, well-documented
Config	Pydantic + PyYAML	Validated, typed configuration
Output	Rich	Progress bars, tables, console formatting

Frontend

Component	Technology	Rationale
Framework	React 18 + TypeScript	Ecosystem, components, talent pool
Build	Vite	Fast builds, modern defaults
UI library	shadcn/ui + Tailwind	Customizable, not opinionated
Code viewer	Monaco Editor	VS Code's renderer, full-featured
Graph viz	Cytoscape.js	Best for code/network graphs
Markdown	react-markdown + rehype	Flexible rendering pipeline
State	Zustand	Simple, fast, minimal boilerplate
Streaming	EventSource API	Native SSE support
Routing	React Router v7	Standard, mature

IDE Extensions

Platform	Technology
VS Code / Cursor	TypeScript, VS Code Extension API
PyCharm / IntelliJ	Kotlin, IntelliJ Platform SDK

DevOps

Component	Technology
Containerization	Docker + docker-compose
CI/CD	GitHub Actions
Package	PyPI (backend), npm (frontend), VS Code Marketplace, JetBrains Marketplace

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17. Non-Functional Requirements

Performance

Metric	Target
AST analysis (1000 files)	< 5 seconds
Chat response (first token)	< 2 seconds
Graph render (5000 nodes)	< 1 second
Incremental update (10 files)	< 30 seconds
Full index (1000 files)	< 30 minutes
UI initial load	< 3 seconds

Scalability

Dimension	Solo Mode	Team Mode
Concurrent users	1	50+
Codebase size	Up to 10K files	Up to 100K files
Graph nodes	Up to 50K	Up to 500K
Conversations	Unlimited	Unlimited

Dimension	Solo Mode	Team Mode
Storage	~100MB per project	Scales with Postgres

Security

Requirement	Implementation
API keys at rest	Encrypted in config, masked in UI
Source code access	Never leaves the server; only docs/summaries sent to LLMs
Auth (team mode)	OAuth 2.0 / API keys / SSO
RBAC	Admin, Member, Viewer roles
Network	HTTPS in production, localhost in solo mode
Audit log	All LLM calls logged with cost and tokens

Reliability

Requirement	Implementation
LLM call retry	Exponential backoff with jitter (3 retries)
Incremental indexing	Hash-based change detection, never re-processes unchanged files
Crash recovery	Pipeline state persisted; resume from last checkpoint
Data backup	SQLite export, graph JSON always on disk

Extensibility

Dimension	Approach
Languages	Plugin system: add parsers for .ts, .js, .go, .rs, .java
LLM providers	litellm supports 100+ providers out of the box
Graph backends	Swappable: NetworkX (local), Neo4j (production)

Dimension	Approach
Vector backends	Swappable: ChromaDB (local), Pinecone, Weaviate
Prompts	External .md files, fully customizable per project
Output formats	Markdown, JSON, HTML, PDF export

Appendix A: Glossary

Term	Definition
Documentation Pyramid	Hierarchical docs: file → directory → module → architecture
Knowledge Graph	Network of code entities and their relationships
RAG	Retrieval-Augmented Generation — feeding relevant docs to an LLM
AST	Abstract Syntax Tree — structured representation of source code
Enrichment	LLM-annotated semantic meaning on graph edges
Rollup	Summarizing child docs into parent-level summaries
Impact Analysis	Tracing what would be affected by a code change
Context Export	Extracting relevant docs/graph for external LLM/agent use

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Document generated: March 22, 2026
Repository: tornadoslims/codebase-cartograph